Self-adaptive Learning Algorithm as a Tool for the Development and Strengthening of the Dyslexic Student's Skills in the Study of Musical Composition



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Abstract In this period, the topic of Artificial Intelligence (AI) is at the center of various discussions that mainly concern the teaching/learning system and the contribution that AI can make in the classroom. From a naturally inclusive perspective, technologies (based on AI) from the early stages of design and development must take into consideration the needs of all students, without neglecting dyslexic students. In that sense, the main problem is represented by the ability of these tools to support the dyslexic student in studying and not in carrying out the assigned tasks for him/her. This research paper presents an algorithm (which combines the Viterbi algorithm with the Markov process) designed and developed to enhance the learning process of a dyslexic student in the area of Music Composition, in particular, in the harmonization of a melodic line. The harmonization of a melodic line involves many cognitive processes and systems and a learning disability may lead to over-fatigue and to a psychological distress of the student, with the subsequent loss of confidence and personal motivation. The algorithm looks like a tutor able to support the student through suggestions during the harmonization of a musical melody (and not by providing the solution). The experimental results show that the algorithm, given a melodic line, is able to correctly generate a bass line as well as harmonize each sound of the melodic line. It is also demonstrated (through a case study) that the algorithm can have a positive impact on dyslexic students' motivation and engagement in learning. Finally, educational implications and research suggestions are provided based on the research results.

Keywords Artificial intelligence · Dyslexia · Educational technology · Hidden Markov Model · Melodic harmonization

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G. Fortino et al. (eds.), *Proceedings of Third International Conference on Computing and Communication Networks*, Lecture Notes in Networks and Systems 917,
https://doi.org/10.1007/978-981-97-0892-5_33

1 Introduction

Nowadays, the introduction of Artificial Intelligence (AI) in the classroom represents a necessary breakthrough. However, this tool must be used with awareness and knowledge of the facts: with proper use, AI can represent a valid and valuable aid for the learning process. The use of technology (based on AI) in the classroom is, however, one of those divisive issues. In fact, it is very often commonly believed that too much technology limits creativity and at the same time offers shortcuts for students (and/or teachers). Instead, it is necessary to become aware of the many possibilities that these tools offer and of the importance they cover, whether we like it or not, in the learning process of students in general but above all of dyslexic students. These tools should not provide the student with the solution to a problem, but help the student find the appropriate solution to the problem. This necessarily implies that AI-based tools must be adequately designed, through a fruitful and fertile integration of multiple strategies that can avoid the creation of educational resources that are by their nature deficient [1]. The growing openness of the Educational Technology World to inclusion can be traced in the trend of the main teaching tools to satisfy the needs of personalized learning [2] where the software should act as an empathic mediator to strengthen motivation, awareness, and mastery of the student's learning process [3]. New technologies must guarantee the acquisition of skills useful for individual growth, and therefore, combine the use of resources and tools with the possibility of developing critical thinking and producing multimedia artifacts from scratch. To enhance the creativity and aptitudes of individual students, it is necessary to stimulate their aptitudes, generating curiosity, and introducing them to new fields of study [4].

In the field of music and more specifically in the area of musical composition, several algorithms have been developed that are able to compose music, using Markov chains [5–7] or neural networks [8–10]. The most recent advances in the field of deep learning have opened the way for a number of new methods for generating music [11–14]. However, all of these systems for music composition focus largely on autonomous generation: little work in this domain has been addressed regarding issues related to the support of the student (and not of the "musician") in educational activities.

In the educational field (in general but more specifically in the music field), the AI should help the student to improve his/her critical thinking [15] and consequently his/her creative thinking [16]: the AI should teach the student to look at things from different perspectives, to test hypotheses, and to find creative solutions to problems. Critical thinking helps the student to make informed decisions, solve problems effectively, and develop independent thinking. This is important for all students but becomes crucial for the dyslexic student.

Starting from these assumptions, this paper presents an algorithm able to support the dyslexic student in harmonizing a melodic line, providing visual suggestions that have the purpose of guiding the student toward the search for a solution: harmonization of each sound of the melodic line avoiding the errors provided by musical

grammar. The algorithm is based on the concept of deep learning, and thanks to the training phase it is able to support the student through the Markov Decision Process.

The structure of this paper has been organized as follows. Section 2 explains the principles behind the harmonization of a melody. Section 3 explores the concept of "dyslexia", which underlies the proposed algorithm. This is followed (Sect. 4) by a description of the method used to achieve the goal. Section 5 shows some experimental evidence illustrating the effectiveness of the proposed method. Finally, in Sect. 6, the paper concludes with concluding remarks on current issues and future research possibilities with respect to the efficient improvement of educational practices and technologies.

2 Melody Harmonization

Along with rhythm and melody, harmony is one of the three constitutive elements of the music. Leaving aside the concept of rhythm which is not the object of this research, the melody can be defined as a succession of sounds with complete meaning (see Fig. 1a), while the harmony represents the simultaneity of two or more sounds built under each sound of the melody (see Fig. 1b) [17].

Harmonizing a melody line means arranging a melody in a harmonic context, submitting a sound texture to the music, and choosing and evaluating the alternatives between the possible chords [17, 18]. The musical chord consists of the superimposition of several sounds that are performed simultaneously (see Fig. 1b). When a chord is arranged in overlapping thirds, it is said to be in the fundamental state (see Fig. 2a): in this case, the lowest note is called fundamental, and the other two notes are called third and fifth (seventh, …). If, on the other hand, the chord is in a position in which the root is not the lowest note, it is said that the chord is in an inverted state: first position and second position (see Fig. 2b). From this, it follows that each sound of the melody can belong to different chords (see Fig. 3).

During his harmonization, the student must understand which chord each sound belongs to on the basis of the previous chord and how the following chords are intended to be developed. When choosing, two important sounds must be taken into consideration: the upper and lower sound of the chord which represents the bass, and the soprano (the lower note gives depth to the chord and the upper one is the most perceptible to the human ear) [18].

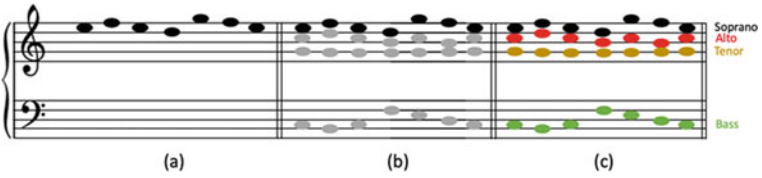


Fig. 1 Example of melody and its harmonization

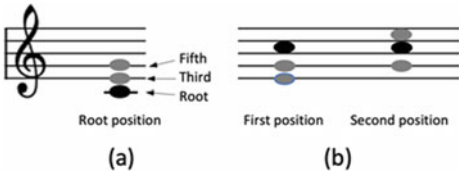


Fig. 2 Example of musical chords



Fig. 3 Examples of possible musical chords for the C note of the melody

For this reason, before harmonizing a melodic line, the student learns to harmonize a bass line (which makes it easier to learn the many musical grammar rules) trying to build an interesting melodic line. For each sound of the bass line, the student is able to precisely define which are the other sounds of the chord (tenor, alto, and soprano) and only has to decide their arrangement to develop the melody (see Fig. 4). Being that the harmonization must be done in 4 voices (see Fig. 1c) and that instead a chord is composed of only 3 sounds, the musical grammar imposes that a sound of the chord is doubled in one of the 4 voices, according to the scale degree of the bass line sound. This offers the possibility to modify the trend of the melodic line (see Fig. 4a and b).

In harmonizing the melody, on the other hand, the student must understand which chord each sound belongs to: it is an important choice that requires time (due to the many possibilities that each sound can have, as seen in Fig. 3) and concentration and which therefore it can lead the student (especially dyslexic) to lose motivation and abandon the exercise.

The algorithm presented in this document intends to help the dyslexic (and non-dyslexic) student in the harmonization of the melody, suggesting (where necessary) which sound can be present in the bass line for each sound of the melody. This allows the student to know the two most important sounds of the chord (bass and soprano) and to apply the previous knowledge acquired during the harmonization of the bass line: the graphical representation (as written in paragraph 3) helps the dyslexic student to memorize certain situations and to know how to recognize and reproduce them when they arise.

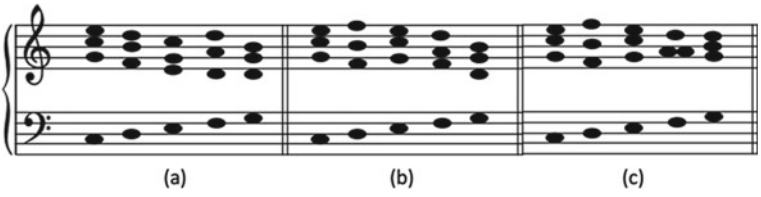


Fig. 4 Example of different bass line harmonization

3 Dyslexia Versus AI (and Vice Versa)

How can AI support the learning process of the dyslexic student?

Dyslexia does not depend on subjective psychological factors, low motivation to study, or low intelligence. Dyslexia consists of the difficulty of automating the process of decoding the graphic signs that characterize the written text [19]. This difficulty has repercussions on the performance which is often characterized by low reading speed and/or poor accuracy and this can compromise the understanding of the text [20]. This may mean difficulties in understanding the rules of musical grammar and consequently little or no practical application of the concepts, i.e., difficulties in harmonizing the sounds of the melody.

The natural compensatory tool of the person with dyslexia is the development of the prediction skill, i.e., "*guessing the words while reading*" [20, 21]. In harmonizing the melody, it is necessary to avoid this instrument becoming the lifeline: harmonization requires a lot of energy and concentration, because it is necessary to identify the sounds of each chord and these depend on the sounds that make up the previous chord and how you intend to build harmony in subsequent chords. Trying to "guess" the sounds of the chord could lead the student to make mistakes (based on musical grammar). Dyslexia involves a reading difficulty that can be compensated if the information is provided through oral channel support (such as an audio message, an audio file, …) [22] or the visual one, for example, using images [23]. It is immediately clear, therefore, how important it is for the dyslexic student the use of study material that is not only present in written mode, but also in graphic mode. In textbooks, the rules of musical grammar are always explained by the written text (rich in specific technical terminology) followed by images representative of the final result that would be obtained from their application. In other words: given some sounds of the melody is proposed an image that shows its harmonization on the basis of the rules explained, and it is up to the student to understand the movement of the sounds of the various voices between the chords that follow. The difficulty of understanding the written text is added to the difficulty of interpreting the image. It goes without saying that it is not enough to simply change the mode of presentation to eliminate any difficulty [24]: it is necessary, in fact, that the student with dyslexia learn to recognize certain melodic situations (for example, a melodic progression, a stylistic feature, …) and to harmonize them (adding to each melody sound the sounds that characterize the musical chord and making the sounds move correctly between one chord and the next).

In order to acquire an effective method of study, the student with dyslexia must be supported [25] by the teacher and by the use of appropriate educational and technological tools called "compensatory tools". Compensatory tools (IT or not) are always aimed at the acquisition of autonomy [22]: they need an initial learning phase to be used at their best and their aim is to improve skills and autonomous learning. Basically, a good compensation tool is one that requires little effort in learning how to use it, that is simple and immediate and that leaves intact, in its user, the resources and energies to decide on the activities of study and understanding of concepts. It should

be stressed that compensatory technologies do not "compensate in themselves". In order for users to be effective, they must also develop in parallel a "compensatory competence", that is, a strategy that allows them to make the best use of the tool and integrate it effectively in the learning phases.

Compensatory tools, based on AI, can play a fundamental role in the implementation of inclusive teaching. Through the use of technologies, the dyslexic student can be allowed to do what he otherwise could not do, or to do it with less expenditure of energy, or to do it in a more independent or safe or psychologically more acceptable way [26].

The algorithm object of this research takes its cue from the above considerations and presents itself as a support tool for the dyslexic student, with the aim of guiding him in the correct harmonization of a melodic line through the suggestion of the possible sounds of the bass line, leaving the completion of the tenor and contralto lines to the student. In this way, the student has the possibility to observe (and memorize) how the movements of the melody and the bass line vary without the interference of other sounds.

4 The Proposed Model

This paragraph illustrates the general architecture of the algorithm designed and developed on the basis of the concepts presented in paragraphs 2 and 3. In summary:

(1) the melodic line and the bass line represent two fundamental elements of the harmony of a musical piece: the trend of the first is closely connected to the second (and vice versa);
(2) the ascending or descending movement of the sounds of the melodic line depends on the movement of the sounds of the bass line (and vice versa): these determine the arrangement of the sounds of the other voices (tenor and alto);
(3) the ascending or descending movement of a sound of the melody depends on the degree of the scale of the sound of the bass line, and therefore, on which sound of the chord is doubled (see example in Fig. 4);
(4) the graphic representation is an important compensatory tool for the dyslexic student: it represents the educational mediator to recall content related to specific concepts [23];
(5) in the harmonization of the melody, the possibility to define the sounds of the bass line with the of the AI prevents the dyslexic student from resorting to the use of the prediction skill and allows him/her to do what he/she otherwise would not could, or to do it with less expenditure of energy, or to do it in a more independent or secure or psychologically more acceptable way: in other words, AI can be a useful tool to motivate the student.

The algorithm starts from a fundamental assumption: *a sound of the melodic line has an ascending or descending movement that depends on the sound of the bass line, which makes it possible to identify the degree of the musical scale to which the*

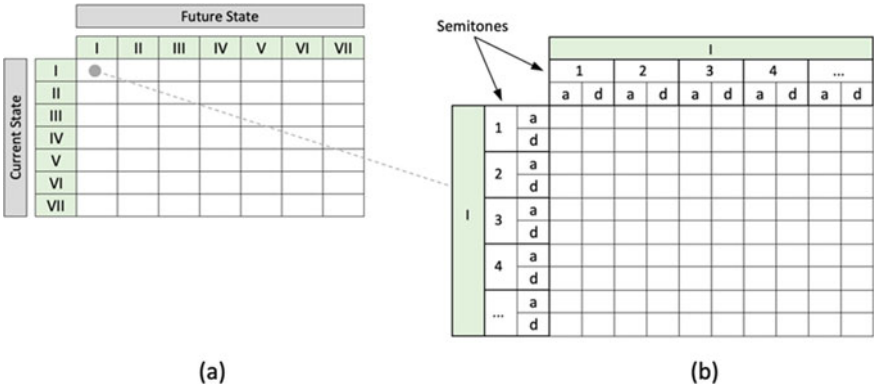


Fig. 5 Example of transition matrix

chord belongs, which in turn defines the sounds of the chord and which ones can be doubled.

This can be represented by a transition matrix P_t that is, for Markov chain $\{X\}$ at time *t,* a matrix containing information on the probability of transitioning between states. In particular, given an ordering of a matrix's rows and columns by the state space S, the (i, j)th element of the matrix P_t is given by

$$(P_t)_{i,j} = P(X_{t+1} = j | X_t = i)$$

For the algorithm, the matrix represents the probability that one degree of the musical scale X_t resolves on another degree of the musical scale (Fig. 5a, column 1 and row 1) [5].

However, this is not enough to describe the movement of the sounds of the melody in the transition from one chord to another. It is necessary to consider simultaneously both the ascending or descending movement of the sound of the melodic line (Fig. 5b, column 3 and row 3) and the number of semitones that characterize this movement (Fig. 5b, column 2 and row 2).

The probabilities calculated on the training corpus are associated with the possible transitions between the various states [27]: starting from any state S_i, the next sound of the melody will be attributable to a given state S_j with a certain probability

$$P(S_i | S_J) = \frac{C(S_i, S_j)}{C(S_i) \cdot N}$$

where $C(S_i, S_j)$ is the count in the corpus of occurrences of the sequence S_i S_j, $C(S_i)$ is the count in the corpus of occurrences of S_i and N is the number of states. All probabilities $P(S_i|S_j)$ are entered into the transition matrix.

If with the transition matrix, it is possible to describe the trend of the melody, to define the bass line (which will determine the degree of the musical scale, and

therefore, allow the identification of the sounds of the chord and their arrangement), it is possible to use the Viterbi algorithm (associated to the transition matrix), which with a dynamic programming approach, allows to analyze the paths between states starting from the initial state going through the successive states identifying the best path in relation to the given emissions [27]. The probability of the most probable path ending in state k with observation i is

$$p_1(i, x) = e_1(i) \max_k (p_k(j, x - 1) \cdot p_{k1})$$

where i represents the probability of observing element i in state l, j represents the probability of the most probable path ending at position $x - 1$ in state k with element j, and p_{kl} represents the probability of the transition from state l to state k.

Training Phase

The use of the algorithm has provided a training phase consisting in the reading of musical pieces composed in the form of a 4-voice choir (soprano, alto, tenor, bass) which included fifth chords (such as C-E–G) and seventh chords (such as C-E–G-B), the inversions of these types of chords and that there were no modulations (i.e., key changes). The musical pieces were in MIDI format: a digital format that does not require complex information decoding operations (therefore, greater speed in the phase of chord analysis and compilation of the transition matrix) and of which it is possible to find a well-stocked database online. The transition matrix did not have any default setting regarding the dimensions, but on the contrary, it was automatically resized during the reading of the chords (based on the type of each chord, the degree of the scale of the chord, the ascending or descending movement of the melody sound in the concatenation of two consecutive chords).

Testing Phase

During the training phase, the algorithm was tested to verify its functionality. In particular, the algorithm was asked to read a melody and propose possible solutions for the bass line. The goal was to look at the algorithm as it was defined and inspect it carefully for imprecisions, so as to be able to identify which melodic passages represented an obstacle for the algorithm, create set tests suitable for verifying the overcoming of those defects and prepare, therefore, an update of the training phase with the reading of musical pieces with specific characteristics.

It emerged that (see Table 1), as the reading of the number of possible cases of concatenation of musical chords increased (training phase):

- the possibility of the bass line not being completed decreased (Table 1 column 2);
- the possibility that the sounds of the bass line represented grades of the scale whose sequence was not correct (according to the rules of musical grammar) decreased (Table 1 column 3);
- the possible solutions for the bass line (corrected from the point of view of musical grammar) increased (Table 1 column 4).

Table 1 Analysis of the algorithm solutions

Number of chords	Uncompleted bass line (%)	Uncorrected grade sequence (%)	Possible solutions (%)
500	96	100	0
1000	87	81	1
1500	72	67	1
2000	59	55	1
3000	31	41	2
4000	28	26	2
5000	24	18	3

As already demonstrated by the various experiences of deep learning, for a better performance of the algorithm, it is necessary to analyze a large number of cases. As regards the present research, it was important to maximize the number of possible solutions of the algorithm (corrected according to the rules of musical grammar) (see Table 1 column 4) to support the student with greater accuracy avoiding loss of motivation, and to give the student interpretative freedom (creativity).

5 Evaluation: A Case Study

The first tests to verify the algorithm were carried out by 3 teachers of music composition, who had the task of verifying the proposed solutions and identifying possible integrations of the training data. Subsequently, a group of students (identified by the 3 teachers) was asked to use the algorithm.

The test was conducted for a time period of 4 months (from January 2023 to April 2023) and involved 12 students, 5 girls, and 7 boys (aged 15–16), of which 6 were affected by dyslexia: all of them had similar educational backgrounds.

During the first month, the students participated in the lessons and carried out the proposed activities, without using the algorithm. The teacher monitored the students' behavior and corrected the exercises by providing suggestions and further explanations (where necessary). At the end of the month, an examination was passed in the classroom consisting of the harmonization of a melody, identical for non-dyslexic and dyslexic students.

The result supplied important information so as to be able to continue with the project. In particular, it emerged that:

- 6 students harmonized all the melody (of which 1 dyslexic student);
- 4 students harmonized about 60% of the melody (of which 3 dyslexic students);
- 2 students (with dyslexia) harmonized less than 40% of the melody.

In February, the teacher introduced the algorithm to the students, who were able to start using it while carrying out exercises.

At the end of the month, a classroom examination was carried out, assigning the same test to all students and allowing them to use the algorithm. The results supplied the hoped-for information:

- 9 students harmonized all the melody (of which 3 dyslexic students);
- 2 students (with dyslexia) harmonized about 90% of the melody;
- 1 student (with dyslexia) harmonized about 80% of the melody.

In the following two months (March and April) the students continued to use the algorithm during the various activities proposed by the teacher, acquiring greater autonomy which was confirmed in the last test proposed at the end of the course: it harmonization of a melody without using the algorithm.

The results were satisfactory:

- 8 students harmonized all the melody (of which 2 dyslexic students);
- 2 students (with dyslexia) harmonized about 85% of the melody;
- 2 students (with dyslexia) harmonized about 70% of the melody.

Beyond the numerical results, one of the important things that emerged concerns the increase in motivation (and therefore self-esteem) of dyslexic students. In fact, it was observed that, at the beginning, if the algorithm stopped because it was unable to give a suggestion, the dyslexic student also stopped; subsequently, the student was able to go ahead with the harmonization, even if only for a few chords. This had tangible feedback on each student's performance, which improved their grades. The learning improvement also appeared for students who already drew a high profit because they learned to find more than one possible solution for the harmonization of the same melody.

6 Discussion and Conclusions

Artificial Intelligence can certainly bring innumerable advantages within the musical composition learning process: on the one hand, it will be essential to educate students to understand AI algorithms (promoting an aware use of the technological platforms that surround them), on the other hand, the same algorithms will facilitate the discovery of many details of the learning processes that are now difficult to detect (therefore useful for the improvement of the algorithms).

It is important to underline two aspects to keep in mind when creating educational software: the centrality of the student in the learning process, and the importance of teacher training, who will not have to be replaced by algorithms, but who will be able to use these tools for an effective teaching activity and deep learning.

In this paper, an algorithm has been presented to support the dyslexic (and non-dyslexic) student in the field of the study of musical composition, which is the harmonization of the melody: a complex and demanding operation that can lead the student to lose motivation, and therefore, to negative results. The simple and

intuitive graphical interface allows the dyslexic student to be able to benefit from a presentation of information through the visual channel.

The results obtained are promising both in terms of the ability of the algorithm to harmonize a melody (or just generate the bass line) and in terms of the motivational aspect of the student (dyslexic and non-dyslexic) which has led to an improvement in educational achievements. Despite this, it is possible to highlight how this type of activity has tried to make the concepts that underlie the learning process experiential and directly tangible at a cognitive and metacognitive level, and therefore, it is necessary to perfect the training phase by reading a greater number of chord concatenation, allowing the algorithm to present alternative solutions for the same melody.

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